

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \multimap A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \multimap PA$ (8.1).

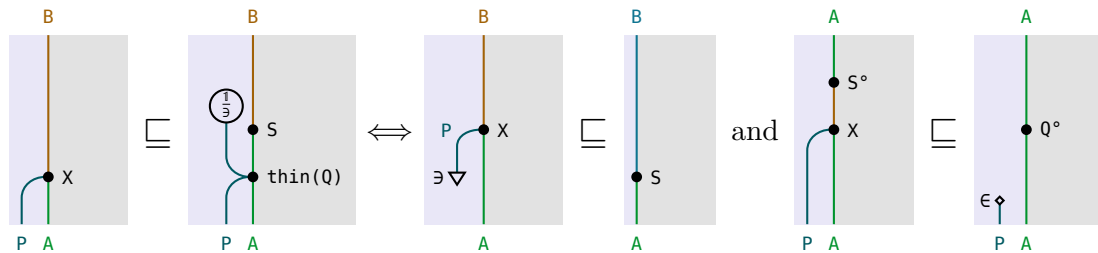
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \multimap ys \multimap xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \multimap b \multimap xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

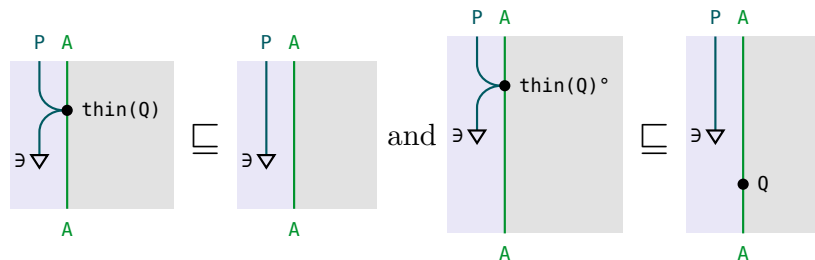
the law	what it says
1 $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$	S is the whole algebra: it produces every candidate that $\text{thin}(Q)$ thins. Everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
2 $\text{thin}(Q) \ni \sqsubseteq \exists$	everything a thinning keeps was in the set
3 $\text{thin}(Q)^\circ \ni \sqsubseteq \exists \ Q$	every element of the set has a Q -lower bound among the kept ones
4 $Q \sqsubseteq R \implies \text{thin}(Q) \sqsubseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
5 $1 \sqsubseteq Q \implies 1 \sqsubseteq \text{thin}(Q)$ $Q \ Q \sqsubseteq Q \implies \text{thin}(Q) \ \text{thin}(Q) \sqsubseteq \text{thin}(Q)$	keeping everything is always a legal thinning
6 $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R) \ \text{preorder}(Q)$ $\implies \text{est}(R) \sqsubseteq \text{thin}(Q) \ \text{est}(R)$	thin-introduction : thinning first cannot lose an R -minimum
7 $\text{thin}(Q) \sqsupseteq \text{est}(Q) \ \frac{1}{\exists} \quad (8.2) - (8.2) \ \text{est}(Q) \ \frac{1}{\exists} \sqsubseteq \text{thin}(Q)$	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
8 $\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \ \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \sqsubseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
9 $P(\text{thin}(Q)) \ \text{union} \sqsubseteq \text{union} \ \text{thin}(Q) \quad (8.4) - (8.4)$ $P(\text{thin}(Q)) \ \text{union} \sqsubseteq \text{union} \ \text{thin}(Q)$	thinning each member set is a thinning of the union

§8.1.1. $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$ — the definition of $\frac{S}{\exists} \text{thin}(Q)$

(8.1.1a)



(8.1.1b)



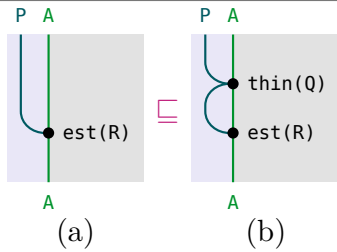
$X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$ at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$, since $\frac{\exists}{\exists} = \Lambda(\exists) = 1$

§8.1.2. $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ — thinning first, then taking the smallest, is the same as taking the smallest

$$\text{preorder}(Q) \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \quad (8.1.2a)$$

est(R)

the $\sqsubseteq$ half: keeping everything is a
thinning — $1 \sqsubseteq Q$



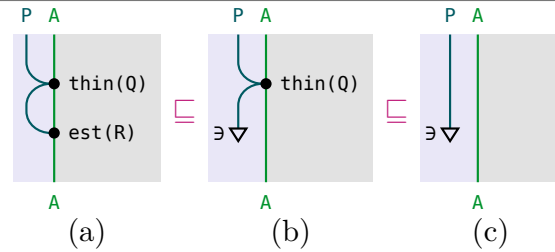
(a)	$\text{est}(R)$	$1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$ $\Rightarrow 1 \sqsubseteq \text{thin}(Q),$ $Q \sqsubseteq Q$ $\Rightarrow \text{thin}(Q)$ $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$
(b)	$\sqsubseteq \text{thin}(Q) \text{ est}(R)$	

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

the $\sqsubseteq$ half, first condition of the UP of

est at $X \triangleq \text{thin}(Q) \text{ est}(R) \text{ — } X \sqsubseteq$

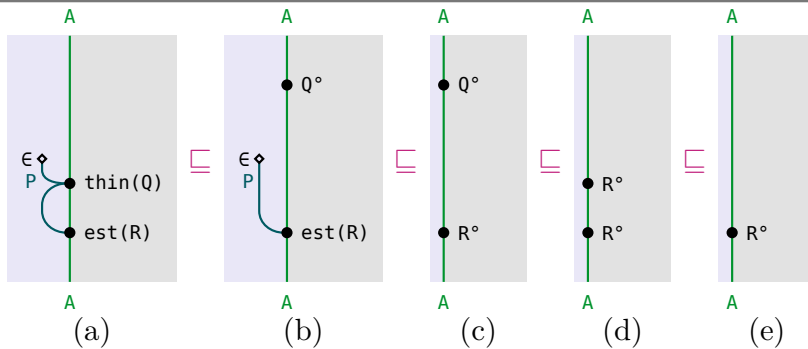
est(R) $\Leftrightarrow X \sqsubseteq \exists \wedge \in X \sqsubseteq R^\circ$



(a)	$\text{thin}(Q) \text{ est}(R)$	$R \cap S \sqsubseteq R$
(b)	$\sqsubseteq \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(c)	$\sqsubseteq \exists$	

$$Q \sqsubseteq R \wedge \text{preorder}(R) \Rightarrow \in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the $\sqsubseteq$ half, second condition



(a)	$\in \text{thin}(Q) \text{ est}(R)$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
(b)	$\sqsubseteq Q^\circ \in \text{est}(R)$	$\in \text{est}(R) \sqsubseteq R^\circ$
(c)	$\sqsubseteq Q^\circ R^\circ$	$Q \sqsubseteq R$
(d)	$\sqsubseteq R^\circ R^\circ$	
(e)	$\sqsubseteq R^\circ$	

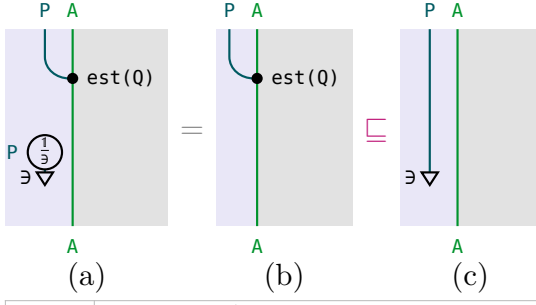
$$\mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq \mathbf{thin}(Q)$$

(8.2)

the singleton holding a Q -least member of a set is a thinning of that set

$$\mathbf{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

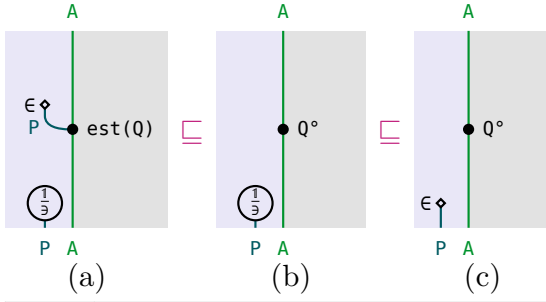
every member of the singleton is a member of the set — $\triangleq \mathbf{thin}, \ni/\ni$ half



(a)	$\mathbf{est}(Q) \stackrel{1}{\exists} \ni$	$\stackrel{1}{\exists} \ni = 1$
(b)	$= \mathbf{est}(Q)$	$R \cap S \sqsubseteq R$
(c)	$\sqsubseteq \ni$	

$$\in \mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in Q to every member of the set — $\triangleq \mathbf{thin}, \in/(Q^\circ \in)$ half

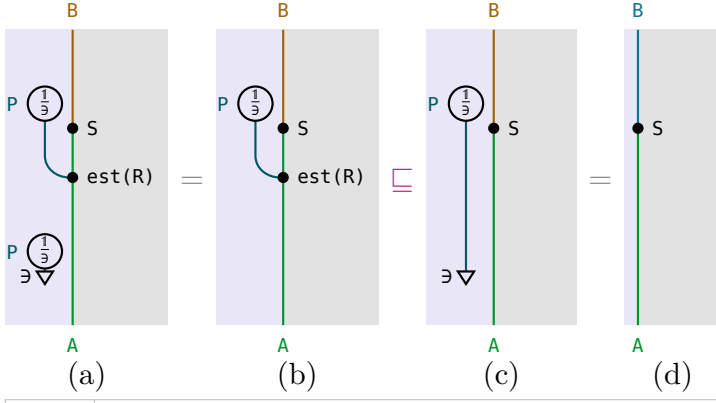


(a)	$\in \mathbf{est}(Q) \stackrel{1}{\exists}$	$\in \mathbf{est}(R) \sqsubseteq R^\circ$
(b)	$\sqsubseteq Q^\circ \stackrel{1}{\exists}$	$\stackrel{1}{\exists} \sqsubseteq \in$
(c)	$\sqsubseteq Q^\circ \in$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq \frac{S}{\exists} \text{thin}(Q) \quad (8.3)$$

$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni \in S$$

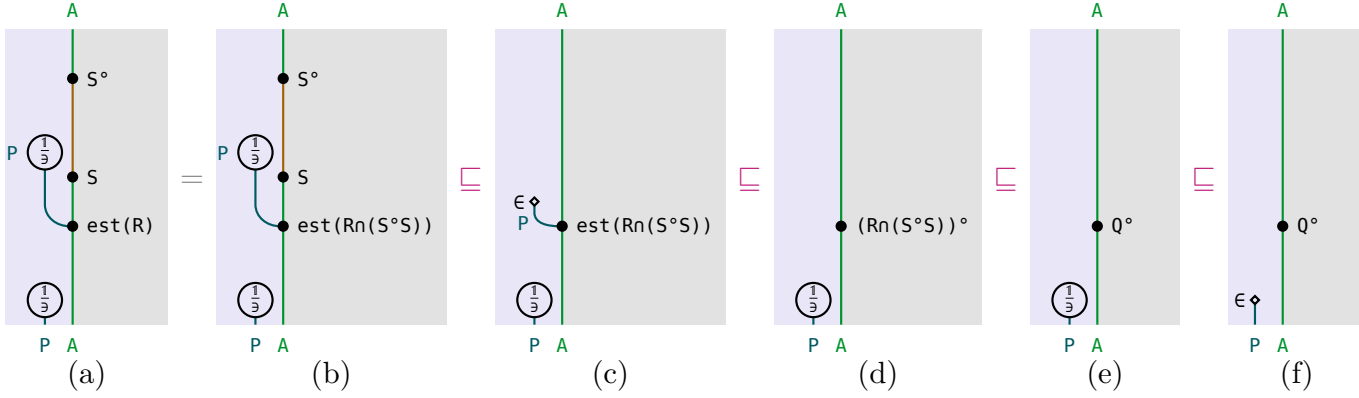
every member of the singleton is a value S returns — $X \subseteq \frac{S}{\exists} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, first condition



(a)	$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni$	$\frac{1}{\exists} \ni = 1$
(b)	$= \frac{S}{\exists} \text{est}(R)$	$R \cap S \subseteq R$
(c)	$\subseteq \frac{1}{\exists} \ni$	$\frac{R}{\exists} \ni = R$
(d)	$= S$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq Q^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument — $X \subseteq \frac{S}{\exists} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, second condition



(a)	$S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$	$\frac{S}{\exists} \text{est}(R \cap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$
(b)	$= S^\circ \frac{S}{\exists} \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$T^\circ \frac{1}{\exists} \subseteq \in$
(c)	$\subseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$\in \text{est}(R) \subseteq R^\circ$
(d)	$\subseteq (R \cap (S^\circ S))^\circ \frac{1}{\exists}$	$R \cap (S^\circ S) \subseteq Q$
(e)	$\subseteq Q^\circ \frac{1}{\exists}$	$\frac{1}{\exists} \subseteq \in$
(f)	$\subseteq Q^\circ \in$	

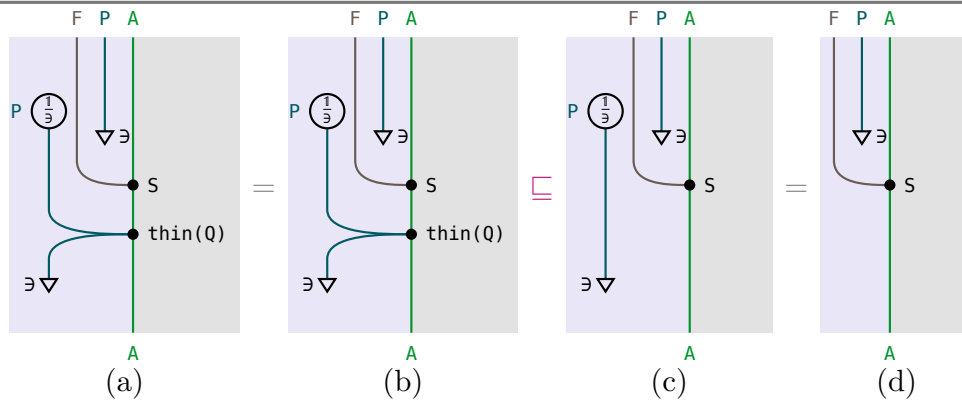
$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq (\frac{S}{\exists} \text{thin}(Q))$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \ni \sqsubseteq (S)$$

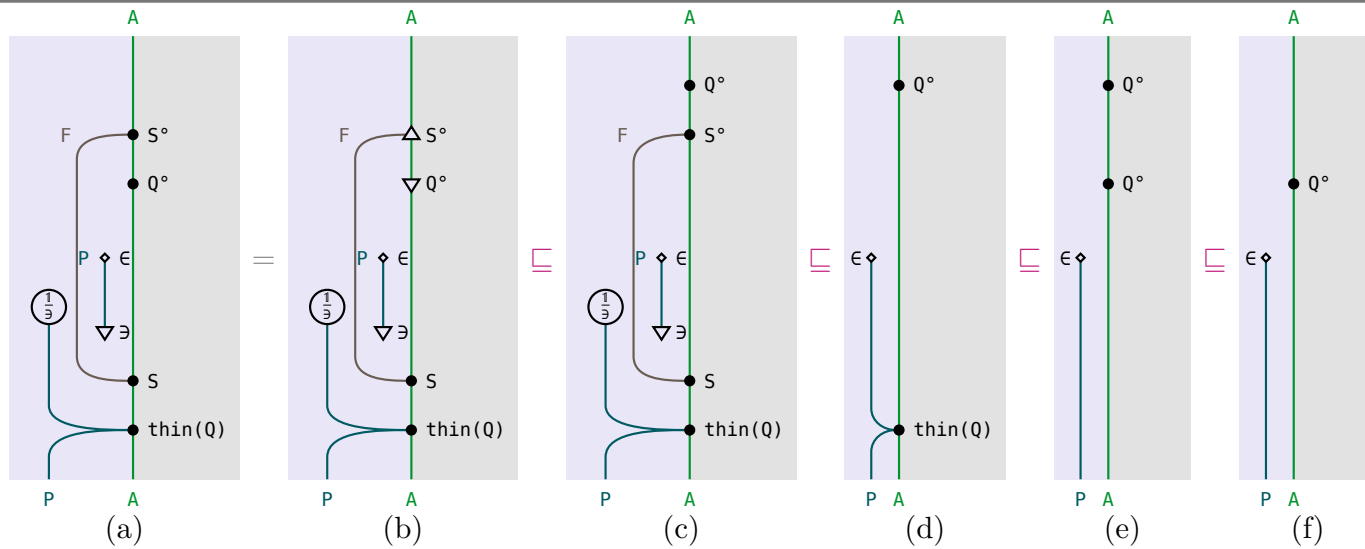
everything the thinning fold keeps is a value of the plain fold — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition, by fusion from the chain below



(a)	$\frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	
(b)	$= \frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	$\text{thin}(Q) \ni \sqsubseteq \exists$
(c)	$\sqsubseteq \frac{F(\exists) S}{\exists} \ni$	$\frac{R}{\exists} \ni = R$
(d)	$= F(\exists) S$	

$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (S)^\circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq Q^\circ \in$$

every value of the plain fold is Q -above one the thinning fold keeps — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition, by hylomorphism theorem from the chain below



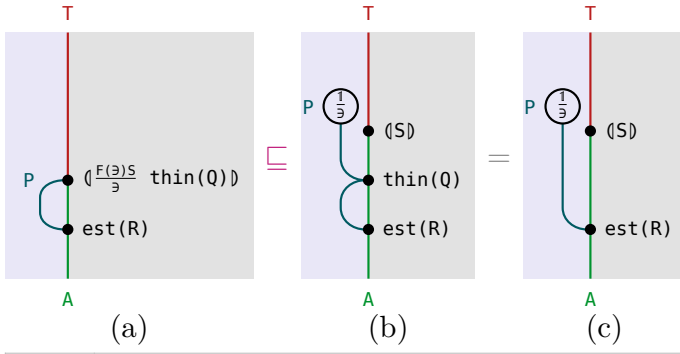
(a)	$S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	
(b)	$= S^\circ F(Q^\circ) F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$F(Q) S \sqsubseteq S Q$
(c)	$\sqsubseteq Q^\circ S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$\implies S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$
(d)	$\sqsubseteq Q^\circ \in \text{thin}(Q)$	$S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \sqsubseteq \epsilon$
(e)	$\sqsubseteq Q^\circ Q^\circ \in$	$\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \in$
(f)	$\sqsubseteq Q^\circ \in$	

(8.1.2d)

	step	type	example	what it does
1	$S:F(A)\rightarrow A$ is 8.1's algebra; example $F(X)=V+V\times X$, $S=[\text{wrap},\text{cons}]$, candidate $a=\text{cons}(v,p)$			
2	S°	$A\rightarrow FA$	$\text{cons}(v,p) \mapsto r(v,p)$	takes a apart into the input of its last step
3	$F(Q^\circ\in)$	$FA\rightarrow F(PA)$	$r(v,p) \mapsto r(v,ps)$, some $p'\in ps$ with $p' \ Q \ p$	replaces each recursive component by a set holding a Q -no-worse one; v stays
4	$\frac{F(\exists)S}{\exists}$	$F(PA)\rightarrow PA$	$r(v,ps) \mapsto \{\text{cons}(v,t) \mid t\in ps\}$	picks one element from each set, combines by S , collects every result
5	$\text{thin}(Q)$	$PA\rightarrow PA$	$\{\text{cons}(v,t) \mid t\in ps\} \mapsto ys$	keeps a subset; every dropped element is Q -beaten by a kept one
6	$\sqsubseteq Q^\circ\in$	$A\rightarrow PA$	ys holds y with $y \ Q \ \text{cons}(v,p') \ Q \ \text{cons}(v,p) = a$	the thinned set still holds an element Q -no-worse than a (S monotonic on Q , Q transitive)

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) S \sqsubseteq S Q \Rightarrow (Q \xrightarrow{F(\exists) S} \text{thin}(Q)) \text{est}(R) \sqsubseteq \frac{(S)}{S} \text{est}(R) \quad \text{Corollary 8.1}$$

the thinning fold refines the optimisation problem itself



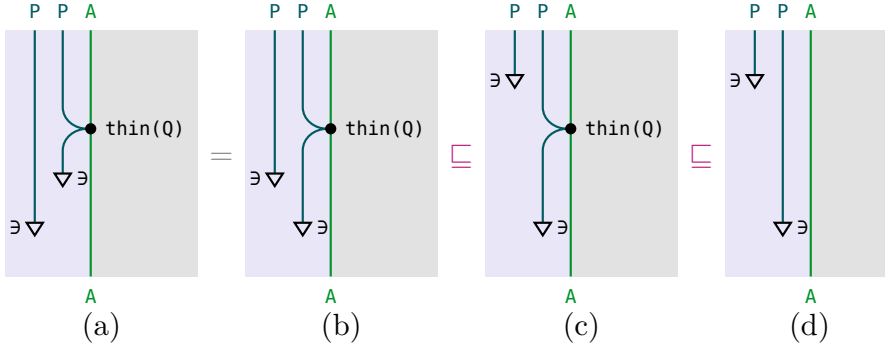
(a)	$(Q \xrightarrow{F(\exists) S} \text{thin}(Q)) \text{est}(R)$	Theorem 8.1 $\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q$ $\Rightarrow (Q \xrightarrow{F(\exists) S} \text{thin}(Q)) \sqsubseteq \frac{(S)}{S} \text{thin}(Q)$
(b)	$\sqsubseteq \frac{(S)}{S} \text{thin}(Q) \text{est}(R)$	$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R)$ $\Rightarrow \text{est}(R) = \text{thin}(Q) \text{est}(R)$
(c)	$= \frac{(S)}{S} \text{est}(R)$	

$$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4)$$

thinning each member set, then taking the union, is a thinning of the union

$$P(\text{thin}(Q)) \text{ union} \exists \sqsubseteq \exists \exists$$

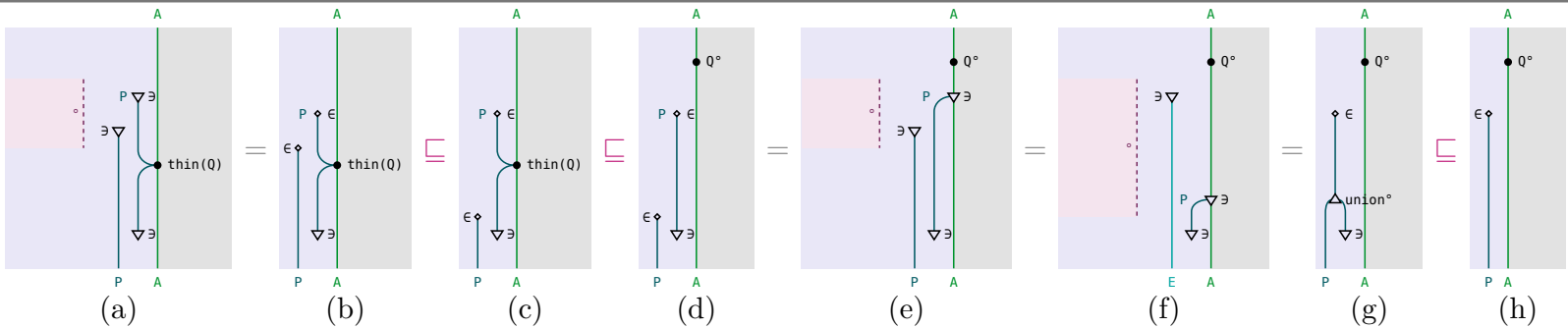
every member of the result is a member of a member set — $X \sqsubseteq \frac{S}{S} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition



(a)	$P(\text{thin}(Q)) \text{ union} \exists$	$\text{union} \exists = \exists \exists$
(b)	$= P(\text{thin}(Q)) \exists \exists$	$P(R) \exists \sqsubseteq \exists R$
(c)	$\sqsubseteq \exists \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(d)	$\sqsubseteq \exists \exists$	

$$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union} \sqsubseteq Q^\circ \in$$

every member of a member set has a Q -lower bound in the result — $X \sqsubseteq \frac{S}{S} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition



(a)	$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union}$	$(R S)^\circ = S^\circ R^\circ$
(b)	$= \in \in P(\text{thin}(Q)) \text{ union}$	$\in P(R) \sqsubseteq R \in$
(c)	$\sqsubseteq \in \text{thin}(Q) \in \text{union}$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
(d)	$\sqsubseteq Q^\circ \in \in \text{union}$	$(R S)^\circ = S^\circ R^\circ$
(e)	$= Q^\circ (\exists \exists)^\circ \text{union}$	$\text{union} \exists = \exists \exists$
(f)	$= Q^\circ (\text{union} \exists)^\circ \text{union}$	$(R S)^\circ = S^\circ R^\circ$
(g)	$= Q^\circ \in \text{union}^\circ \text{union}$	$\text{union} = E(\exists)$
(h)	$\sqsubseteq Q^\circ \in$	

§8.2. Paths in a layered network

(8.2a)

	name	type	definition	meaning
1	F	$\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$	$F(B, X) = B + (B \times X)$	a path is either one vertex a , or a vertex a followed by a path xs
2	α	$F(A, LA) \rightarrow LA$	$\alpha = [\text{wrap}, \text{cons}]$	build a path: wrap one vertex as $[a]$, or put a in front of xs
3	wrapz	$A \rightarrow LA \times \mathbb{N}$	$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle, a \mapsto ([a], 0)$	a path of one vertex, no edge, cost 0
4	consw	$A \times (LA \times \mathbb{N}) \rightarrow LA \times \mathbb{N}$	$(a, (xs, n)) \mapsto ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$	put a in front of the path; the cost adds the weight of the new edge $a \rightarrow \text{head}(xs)$
5	cost	$LA \rightarrow \mathbb{N}$	$\text{cost} \triangleq \llbracket [\text{wrapz}, \text{consw}] \rrbracket \pi_2$	the cost of $[a_0, \dots, a_n]$ is $\text{wt}(a_0, a_1) + \dots + \text{wt}(a_{n-1}, a_n)$
6	R	$LA \rightarrow LA$	$R \triangleq \text{cost} \leq \text{cost}^\circ$	xs is cheaper than ys
7	$L(R)$	$LA \rightarrow LB$	$\llbracket F(\exists, 1) \alpha \rrbracket = L(\exists)$	$L(\exists)$ relates a list of layers to every path that chooses one vertex from each layer
8	minpath	$L(PA) \rightarrow LA$	$\text{minpath} \sqsubseteq \frac{L(\exists)}{\exists} \text{est}(R)$	the input $[x_0, \dots, x_n] : L(PA)$ is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and minpath returns a cheapest one
9	Q	$LA \rightarrow LA$	$Q \triangleq R \cap (\text{head} \text{ head}^\circ)$	like R , and the two paths also start at the same vertex: $\text{head}(xs) = \text{head}(ys)$
10	l	$B \rightarrow B + C$	$a \mapsto l(a)$	mark a value as the left alternative
11	r	$C \rightarrow B + C$	$b \mapsto r(b)$	mark a value as the right alternative
12	cpl	$PB \times A \rightarrow P(B \times A)$	$\text{cpl} \triangleq \text{cp}(1 \times A)$	pair every element of the set with the value beside it
13	cpr	$A \times PB \rightarrow P(A \times B)$	$\text{cpr} \triangleq \text{cp}(A \times 1)$	pair the value beside the set with every element of the set
14	$\frac{F(\exists, 1)}{\exists}$	$F(PB, X) \rightarrow PF(B, X)$	$\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$	cpl transpose: from layer x and a set of paths ps , choose a vertex $a \in x$
15	$\frac{F(1, \exists)}{\exists}$	$F(B, PX) \rightarrow PF(B, X)$	$\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$	cpr transpose: from a vertex a and a set of paths ps , choose a tail $xs \in ps$
16	step	$A \times P(LA) \rightarrow LA$	$\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$	put a in front of every path in ps and keep a cheapest one

(8.2b)

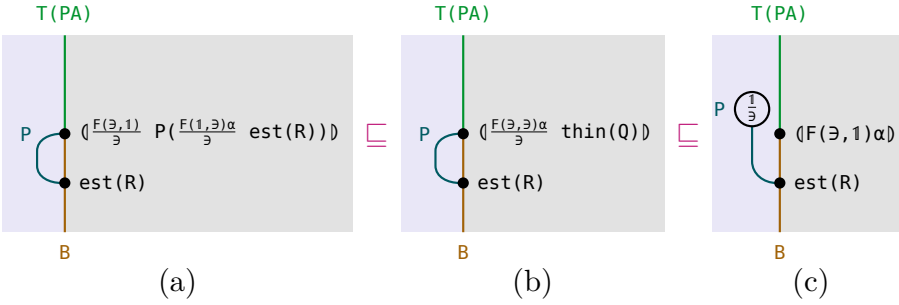
	the law	what it says
1	$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
2	$S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(1, Q) F(\exists, 1) \alpha \sqsubseteq F(\exists, 1) \alpha Q \wedge R \sqcap ((F(1, \exists) \alpha)^\circ F(1, \exists) \alpha) \quad (8.2c)$$

$$\sqsubseteq Q \Rightarrow \langle \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \rangle \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers

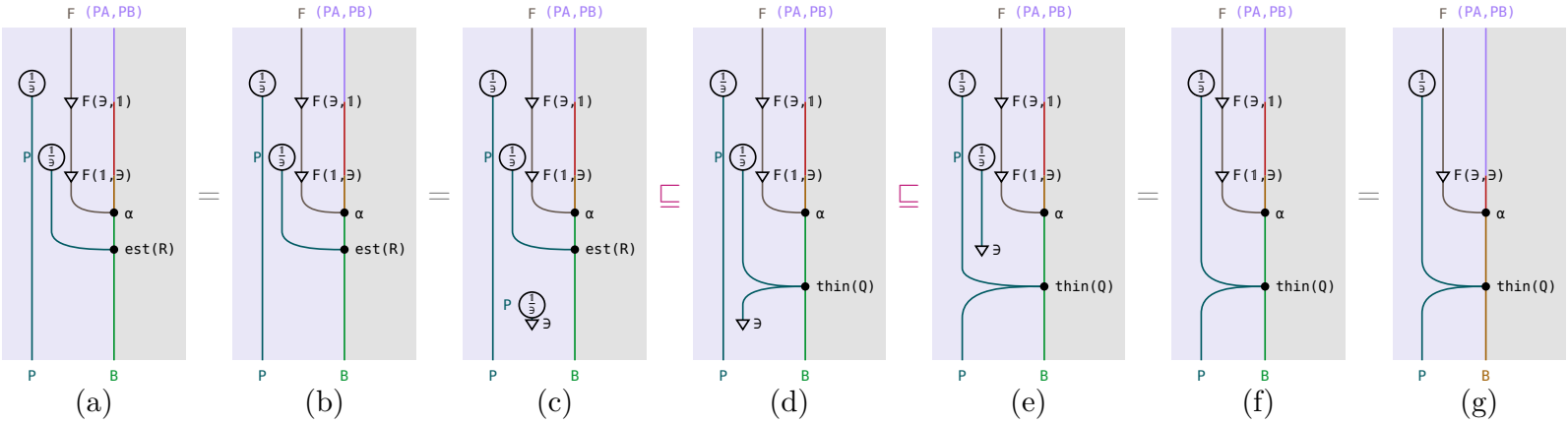
- Why it is needed: $\text{est}(R)$ keeps one cheapest w of the set $\Lambda(S)(x)$; $\text{thin}(Q)$ needs every dropped z to be Q -beaten by a kept one. For $\{w\}$ alone to be a thinning we need $w Q z$ for every z in $\Lambda(S)(x)$.
- We have $w R z$ (w is cheapest) and $w (S^\circ S) z$ (both are built by S from the same argument x); the hypothesis turns the two into $w Q z$: among candidates from one argument, cheaper already means Q -better.



(a)	$\langle \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \rangle \text{est}(R)$	(8.2d) $R \sqcap (S^\circ S) \sqsubseteq Q$ $\Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$
(b)	$\sqsubseteq \langle \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \rangle \text{est}(R)$	$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(1, Q)$ $F(\exists, 1) \alpha \sqsubseteq F(\exists, 1) \alpha Q$
(c)	$\sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \text{est}(R)$	$\Rightarrow \langle \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \text{est}(R)$

$$S \triangleq F(1, \exists) \alpha, R \sqcap (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases



(a)	$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R))$	
(b)	$= \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R))$	$P(\frac{1}{3}) E(\exists) = 1$
(c)	$= \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \frac{1}{3}) E(\exists)$	(8.3) $R \sqcap (S^\circ S) \sqsubseteq Q$ $\Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{3} \sqsubseteq \frac{S}{\exists} \text{thin}(Q)$
(d)	$\sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists}) P(\text{thin}(Q)) E(\exists)$	(8.4) $P(\text{thin}(Q)) \text{union} \sqsubseteq \text{union thin}(Q)$
(e)	$\sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists}) E(\exists) \text{thin}(Q)$	$\frac{V S}{\exists} = \frac{V}{\exists} P(\frac{S}{\exists}) \text{union}$
(f)	$= \frac{F(\exists, 1) F(1, \exists) \alpha}{\exists} \text{thin}(Q)$	$F(R, 1) F(1, S) = F(R, S)$
(g)	$= \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$	

$$= \text{cp}(F) P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}]$$

at the layered network $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}])$: $\frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] - \text{step} \triangleq \text{cpr}$
 $P(\text{cons}) \text{est}(R)$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} - \frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$$

$$\begin{array}{ccccc} F(A, X) & = & A & + & A \times X \\ & & | & & | \\ & & R & & S \\ & & \downarrow & & \downarrow \\ F(A', X') & = & A' & + & A' \times X' \end{array}$$

(8.2e)

u $F(R, S)$ v holds iff one of:

- u, v are both inl and $d R d'$: the leaf moves by R;
- u, v are both inr and $p_1 R q_1 \wedge p_2 S q_2$: the vertex moves by R, the rest by S.
- An inl never relates to an inr : $R + S \triangleq [R \text{ inl}, S \text{ inr}]$ returns to the summand it came from, and $F(1) = 1$ allows no crossing.
- $F(\exists, \exists)$ relates $\text{inr}(\{a, b\}, \{p, q\})$ to $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$, and $\text{inl}(\{a, b\})$ to $\text{inl}(a), \text{inl}(b)$.

§8.3. Implementing thin

(8.3a)

setify : $[A] \rightarrow PA$, $\text{cup} : PA \times PA \rightarrow PA$, $\text{cp}(F) \text{ cp}(F) \triangleq \frac{F(\exists)}{3}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$ for $\leq$ a connected preorder.

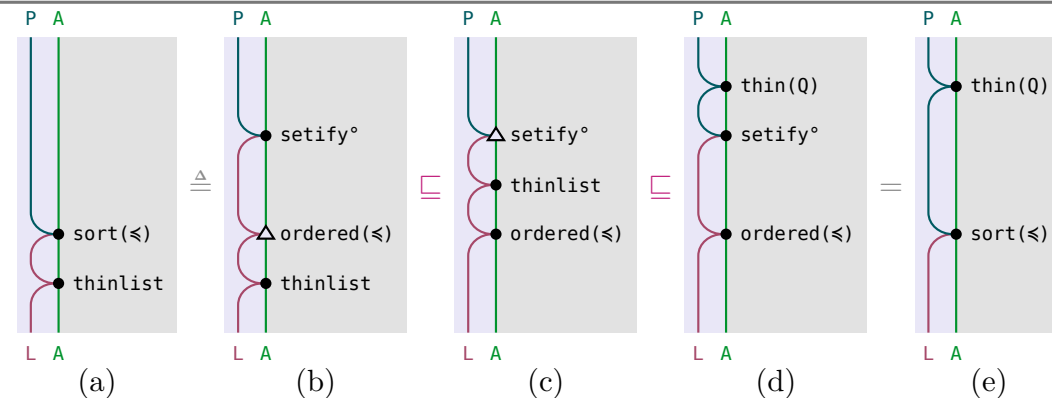
$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $Q[\text{nil}, \text{bump } Q]$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; $\leq$ a connected preorder with both f_1, f_2 monotonic on $\leq$; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

(8.3b)

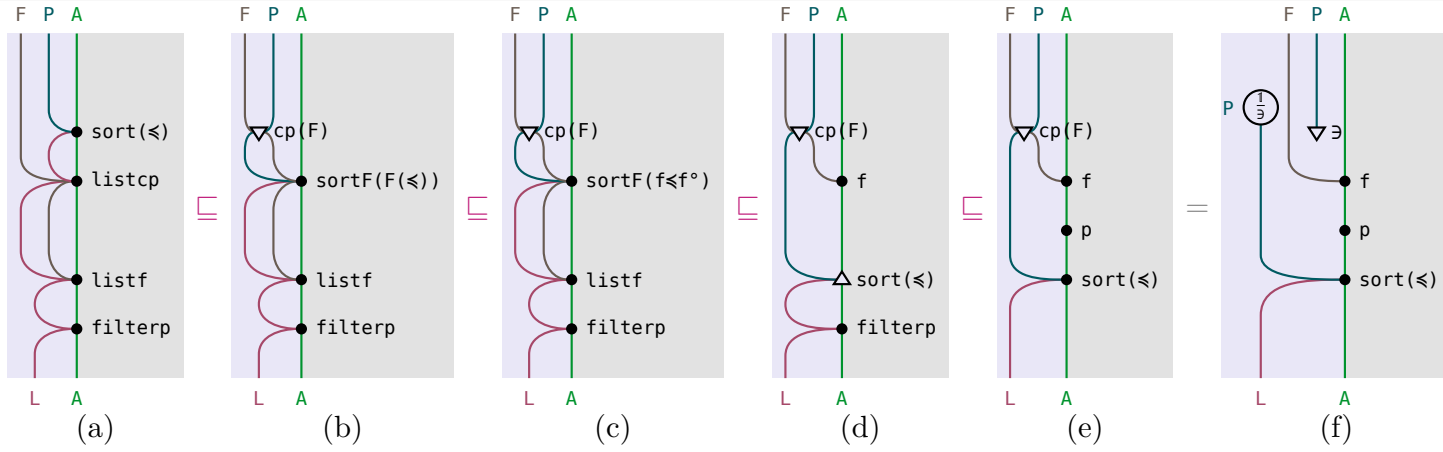
the law	what it says
1 $\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
2 $\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ (8.6) — (8.6) $\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
3 $\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{minlist} \sqsubseteq \text{setify} \exists \wedge (\text{setify} \exists)^\circ \text{minlist} \sqsubseteq R^\circ \Rightarrow \text{sort}(\leq) \text{ minlist} \sqsubseteq \text{est}(R)$ (8.7)	a minimum of the sorted list is a minimum of the set
4 $\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq \text{P}(f) \text{ sort}(\leq)$ (8.8)	shunt a function through a sort
5 $\text{sort}(\leq) \text{ filter}(p) \sqsubseteq \text{E}(p) \text{ sort}(\leq)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
6 $(\text{setify}^\circ \times \text{setify}^\circ) \text{ merge}(\leq) \sqsubseteq \text{cup setify}^\circ \wedge (\text{ordered}(\leq) \times \text{ordered}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{merge}(\leq) \text{ ordered}(\leq) \Rightarrow (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)	merging two sorted lists sorts their union
7 $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

$\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ (8.6)
a thinning of the sorted list lists a thinning of the set — $\leq$ a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.



(a)	$\text{sort}(\leq) \text{ thinlist}$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$
(b)	$\triangleq \text{setify}^\circ \text{ ordered}(\leq) \text{ thinlist}$	$\text{Coreflexive}(\text{ordered}(\leq)) \wedge g \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \Rightarrow \text{setify}^\circ \text{ ordered}(\leq) g \sqsubseteq \text{setify}^\circ g \text{ ordered}(\leq)$
(c)	$\sqsubseteq \text{setify}^\circ \text{ thinlist ordered}(\leq)$	$\text{Map}(\text{setify}) \wedge g \text{ setify} \sqsubseteq \text{setify } T \Rightarrow \text{setify}^\circ g \text{ ordered}(\leq) \sqsubseteq T \text{ setify}^\circ \text{ ordered}(\leq)$
(d)	$\sqsubseteq \text{thin}(Q) \text{ setify}^\circ \text{ ordered}(\leq)$	
(e)	$= \text{thin}(Q) \text{ sort}(\leq)$	

Map(f) $\wedge$ ($\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$) $\wedge F(\leq) f \sqsubseteq f \leq \wedge \text{sortF}(f \leq f^\circ) \text{ listf} \sqsubseteq$ Lemma 8.1
 $P(f) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp} \sqsubseteq E(p) \text{ sort}(\leq) \wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq))$
 $\Rightarrow F(\text{sort}(\leq)) \text{ listcp} \text{ listf} \text{ filterp} \sqsubseteq \frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$
 one sorted list built from sorted arguments, instead of a set built and then sorted — $f : \mathbf{FA} \rightarrow \mathbf{A}$ monotonic
 on $\leq$, p coreflexive, F linear.



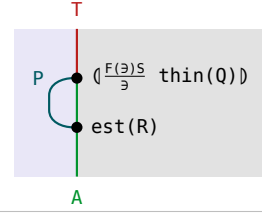
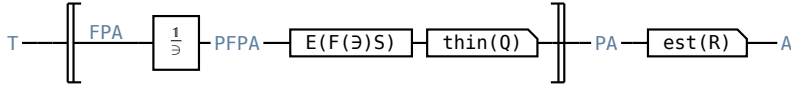
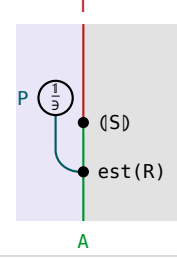
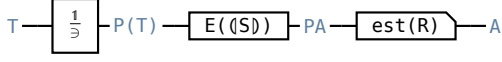
(a)	$F(\text{sort}(\leq)) \text{ listcp} \text{ listf} \text{ filterp}$	$F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq))$
(b)	$\sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \text{ listf} \text{ filterp}$	$\text{Map}(f)$
(c)	$\sqsubseteq \text{cp}(F) \text{ sortF}(f \leq f^\circ) \text{ listf} \text{ filterp}$	$\Rightarrow F(R) f \sqsubseteq f R \iff F(R) \sqsubseteq f R f^\circ$
(d)	$\sqsubseteq \text{cp}(F) P(f) \text{ sort}(\leq) \text{ filterp}$	$\text{sortF}(f \leq f^\circ) \text{ listf} \sqsubseteq P(f) \text{ sort}(\leq)$
(e)	$\sqsubseteq \text{cp}(F) E(f p) \text{ sort}(\leq)$	$\text{sort}(\leq) \text{ filterp} \sqsubseteq E(p) \text{ sort}(\leq)$
(f)	$= \frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$	$\text{Map}(f)$
		$\Rightarrow P(f) = E(f)$
		$\frac{S}{\exists} E(R) = \frac{S R}{\exists}$

Theorem 8.2 $\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) \quad f_1 \ p_1 \sqsubseteq f_1 \ p_1 \ Q \wedge F(Q) \quad f_2 \ p_2 \sqsubseteq f_2 \ p_2 \ Q \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge F(\leq) \quad f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \quad f_2 \sqsubseteq f_2 \leq \wedge \text{sortF}(f_1 \leq f_1^\circ) \quad \text{listf}_1 \sqsubseteq P(f_1) \quad \text{sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \quad \text{listf}_2 \sqsubseteq P(f_2) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_1 \sqsubseteq E(p_1) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_2 \sqsubseteq E(p_2) \quad \text{sort}(\leq) \wedge F(\text{sort}(\leq)) \quad \text{listcp} \sqsubseteq \text{cp}(F) \quad \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \quad \text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1 \quad \text{filterp}_1 \wedge g_2 = \text{listf}_2 \quad \text{filterp}_2 \wedge S = (f_1 \ p_1) \cup (f_2 \ p_2) \Rightarrow (\text{listcp} \langle g_1, g_2 \rangle \text{merge}(\leq) \text{thinlist}(Q)) \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \text{est}(R)$

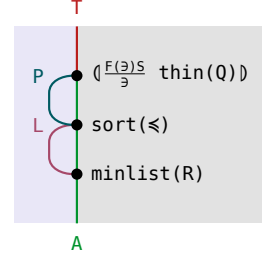
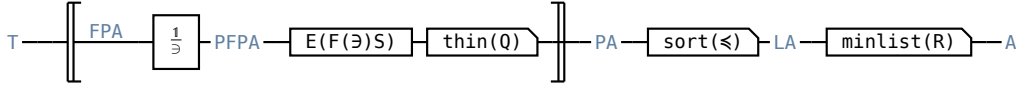
a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

circuit

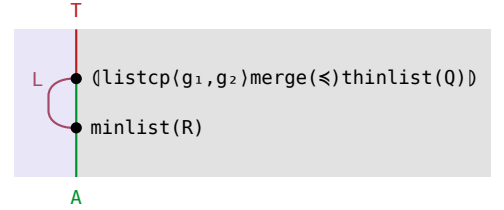
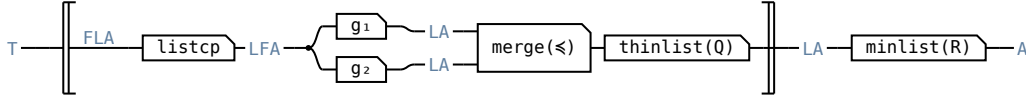
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Corollary 8.1 $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) \quad S \sqsubseteq S \quad Q \Rightarrow (\frac{F(3)S}{3} \text{thin}(Q)) \text{est}(R) \sqsubseteq \frac{(S)}{3} \text{est}(R)$ at $f_1 p_1$ and $f_2 p_2$ monotonic on Q — $\triangleq \text{thinlist}$



$\text{sort}(\leq) \text{minlist}(R) \sqsubseteq \text{est}(R) \text{ — Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{minlist} \sqsubseteq \text{setify} \circ \wedge (\text{setify} \circ)^\circ \text{minlist} \sqsubseteq R^\circ \Rightarrow \text{sort}(\leq) \text{minlist} \sqsubseteq \text{est}(R)$ at its $Q \triangleq R$



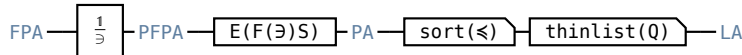
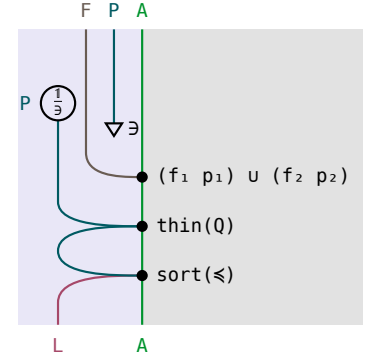
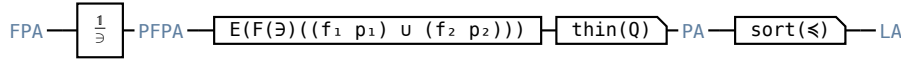
fusion at $\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge F(\leq) \quad f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \quad f_2 \sqsubseteq f_2 \leq \wedge \text{sortF}(f_1 \leq f_1^\circ) \quad \text{listf}_1 \sqsubseteq P(f_1) \quad \text{sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \quad \text{listf}_2 \sqsubseteq P(f_2) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_1 \sqsubseteq E(p_1) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_2 \sqsubseteq E(p_2) \quad \text{sort}(\leq) \wedge F(\text{sort}(\leq)) \quad \text{listcp} \sqsubseteq \text{cp}(F) \quad \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \quad \text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \quad \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \quad \text{listcp} \langle \text{listf}_1 \text{filterp}_1, \text{listf}_2 \text{filterp}_2 \rangle \text{merge}(\leq) \text{thinlist}(Q) \sqsubseteq \frac{F(3) ((f_1 \ p_1) \cup (f_2 \ p_2))}{3} \text{thin}(Q) \text{sort}(\leq)$

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge F(\leq) f_1 \sqsubseteq f_1 \leq \wedge F(\leq) f_2 \sqsubseteq f_2 \leq \quad (8.3c) \\
& \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{listf}_1 \sqsubseteq P(f_1) \text{sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{listf}_2 \sqsubseteq P(f_2) \text{sort}(\leq) \wedge \text{sort}(\leq) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{sort}(\leq) \wedge \text{sort}(\leq) \text{filterp}_2 \sqsubseteq E(p_2) \text{sort}(\leq) \wedge F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \sqsubseteq \text{cup sort}(\leq) \wedge \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \text{listcp} (\text{listf}_1 \text{filterp}_1, \text{listf}_2 \text{filterp}_2) \text{merge}(\leq) \text{thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{thin}(Q) \text{sort}(\leq)
\end{aligned}$$

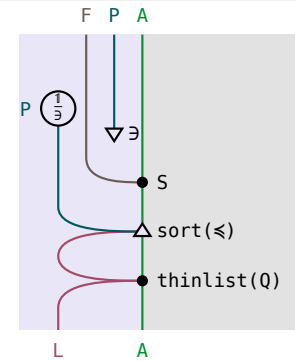
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

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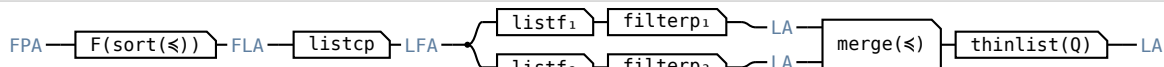
$$\begin{aligned}
& \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq) - \text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \\
& \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist} \\
& \text{setify} \sqsubseteq \text{setify thin}(Q) \\
& \Rightarrow \text{sort}(\leq) \text{thinlist} \sqsubseteq \text{thin}(Q) \text{sort}(\leq)
\end{aligned}$$



$$\begin{aligned}
& \text{FPA} \rightarrow \left[\begin{array}{l} \text{[1/3]} \\ \text{PFPA} \end{array} \right] \rightarrow \left[\begin{array}{l} E(F(\exists) f_1 p_1) \\ E(F(\exists) f_2 p_2) \end{array} \right] \rightarrow \text{PA} \rightarrow \text{cup} \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left(\frac{F(\exists) f_1 p_1}{\exists}, \frac{F(\exists) f_2 p_2}{\exists} \right) \text{cup sort}(\leq) \text{thinlist}(Q) \\
& S = (f_1 p_1) \cup (f_2 p_2) - \triangle \text{thinlist, then } \triangle \text{cup}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow \left[\begin{array}{l} \text{[1/3]} \\ \text{PFPA} \end{array} \right] \rightarrow \left[\begin{array}{l} E(F(\exists) f_1 p_1) \\ E(F(\exists) f_2 p_2) \end{array} \right] \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \left(\frac{F(\exists) f_1 p_1}{\exists}, \frac{F(\exists) f_2 p_2}{\exists} \right) (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \text{thinlist}(Q) \\
& (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \sqsubseteq \text{cup sort}(\leq) - (\text{setify}^\circ \times \text{setify}^\circ) \text{merge}(\leq) \sqsubseteq \text{cup} \\
& \text{setify}^\circ \wedge (\text{ordered}(\leq) \times \text{ordered}(\leq)) \text{merge}(\leq) \sqsubseteq \text{merge}(\leq) \text{ordered}(\leq) \\
& \Rightarrow (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \sqsubseteq \text{cup sort}(\leq)
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow \left[\begin{array}{l} \text{[1/3]} \\ \text{PFPA} \end{array} \right] \rightarrow \left[\begin{array}{l} E(F(\exists) f_1 p_1) \\ E(F(\exists) f_2 p_2) \end{array} \right] \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left(\frac{F(\exists) f_1 p_1}{\exists} \text{sort}(\leq), \frac{F(\exists) f_2 p_2}{\exists} \text{sort}(\leq) \right) \text{merge}(\leq) \text{thinlist}(Q) \\
& (X, Y) (\text{sort}(\leq) \times \text{sort}(\leq)) = (X \text{sort}(\leq), Y \text{sort}(\leq)) - \text{absorption}
\end{aligned}$$



$$F(\text{sort}(\leq)) \text{listcp} (g_1, g_2) \text{merge}(\leq) \text{thinlist}(Q)$$

Lemma 8.1 $\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge F(\leq) f \sqsubseteq f \leq \wedge$

$$\begin{aligned}
& \text{sortF}(f \leq f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(\leq) \wedge \text{sort}(\leq) \text{filterp} \sqsubseteq E(p) \text{sort}(\leq) \wedge \\
& F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\leq))
\end{aligned}$$

$$\begin{aligned}
& \Rightarrow F(\text{sort}(\leq)) \text{listcp listf filterp} \sqsubseteq \frac{F(\exists) f p}{\exists} \text{sort}(\leq) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then} \\
& X(g_1, g_2) \sqsubseteq (Xg_1, Xg_2) - X \langle R, S \rangle \sqsubseteq \langle X R, X S \rangle ; g_i \triangleq \text{list}(f_i) \text{filter}(p_i) - \triangleq \\
& \text{thinlist}
\end{aligned}$$

§8.4. The knapsack problem

(8.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}$, $\text{value} \triangleq \text{list}(\text{vol}) \text{ sum}$, $\text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}$.

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2])$, within w the coreflexive on xs with weight $xs \leq w$, $0 \leq w$.

$R \triangleq \text{value} \geq \text{value}^\circ$, $Q \triangleq \text{Rn}(\text{weight} \leq \text{weight}^\circ)$, $P \triangleq R$,.

$\text{FA} = 1 + \text{Item} \times A$, $\text{listcp} = \text{wrap} + \text{cpr}$, $g_1 \triangleq \text{list}([\text{nil}, \text{cons}])$ $\text{filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$,
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2]$.

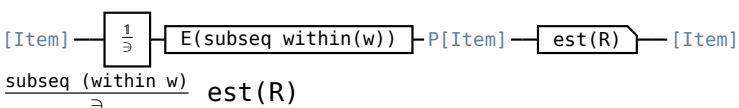
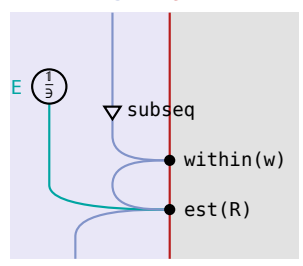
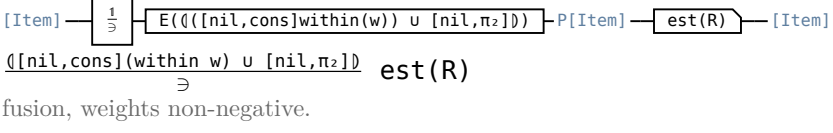
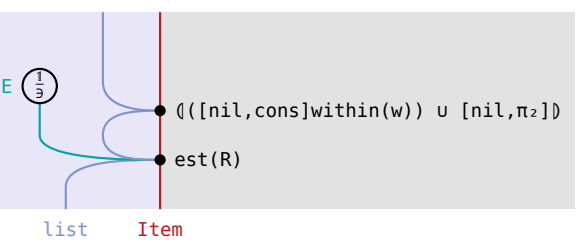
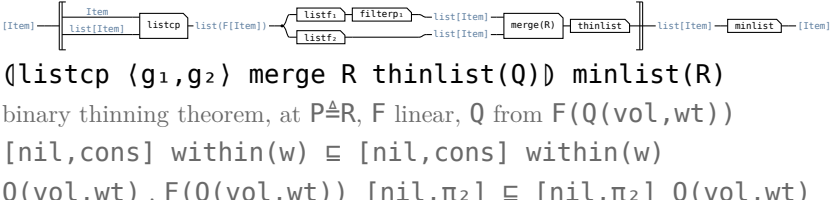
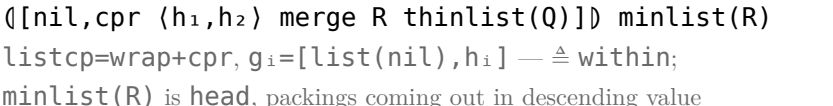
$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$, $h_2 \triangleq \text{list}(\pi_2)$.

(8.4b)

the law	what it says
1 $(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R : a selection of greater value need not still fit once one more item goes in
2 $F(Q(\text{vol}, \text{wt})) [\text{nil}, \text{cons}] \text{ within}(w) \sqsubseteq [\text{nil}, \text{cons}]$ $\text{within}(w) Q(\text{vol}, \text{wt})$ $F(Q(\text{vol}, \text{wt})) [\text{nil}, \pi_2] \sqsubseteq [\text{nil}, \pi_2] Q(\text{vol}, \text{wt})$,	both halves are monotonic on Q once ties in value are broken by weight

$0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{nil}, \text{cons}] R$ (8.4c)
 $[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sort}(R) \wedge \text{sortF}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq$
 $P([\text{nil}, \pi_2]) \text{sort}(R) \wedge \text{sort}(R) \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sort}(R) \wedge F(\text{sort}(R)) \text{listcp} \sqsubseteq \text{cp}(F)$
 $\text{sortF}(F(R)) \wedge (\text{sort}(R) \times \text{sort}(R)) \text{merge}(R) \sqsubseteq \text{cup } \text{sort}(R) \wedge \text{sort}(R) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt}))$
 $\text{sort}(R) \wedge \text{sort}(R) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp } (\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{merge}(R) \text{thinlist})$
 $\text{minlist} \sqsubseteq \frac{\text{subseq } \text{within}(w)}{\supset} \text{est}(R)$

the knapsack problem, as a fold that thins the packings kept at each item

circuit	Hinze–Marsden
 $\text{subseq } (\text{within } w) \supset \text{est}(R)$	
 fusion, weights non-negative.	
 $\text{listcp } (g_1, g_2) \text{ merge } R \text{ thinlist}(Q) \text{ minlist}(R)$ $\sqsubseteq$ binary thinning theorem, at $P \triangleq R$, F linear, Q from $F(Q(\text{vol}, \text{wt}))$ $[\text{nil}, \text{cons}] \text{ within}(w) \sqsubseteq [\text{nil}, \text{cons}] \text{ within}(w)$ $Q(\text{vol}, \text{wt}), F(Q(\text{vol}, \text{wt})) [\text{nil}, \pi_2] \sqsubseteq [\text{nil}, \pi_2] Q(\text{vol}, \text{wt})$	
 $\text{listcp} = \text{wrap} + \text{cpr}$, $g_1 = [\text{list}(\text{nil}), h_1] \text{ — } \triangleq \text{within}$; $\text{minlist}(R)$ is head, packings coming out in descending value	

(8.4a)

§8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

the law	what it says
1 $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
2 $F(Q(\text{len},w)) \text{ [wrap wrap,new]} \sqsubseteq \text{[wrap wrap,new]} Q(\text{len},w)$ $(\forall a. 0 \leq \text{len}(a)) \implies F(Q(\text{len},w)) \text{ [wrap wrap,glue]}$ $\text{ok}(w) \sqsubseteq \text{[wrap wrap,glue]} \text{ ok}(w) Q(\text{len},w)$ $\text{cons monotonic on } \text{collect} \leq \text{collect}^\circ.$	both halves are monotonic on Q once ties in waste are broken by the first line
3 $\text{merge}(\tau) = \text{cat}$	τ needs no sorting at all

$$\begin{aligned}
& (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{new}] \tau [\text{wrap} \text{ wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap} \text{ wrap}, \text{new}]) \text{sort}(\tau) \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{glue}] \tau [\text{wrap} \text{ wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap} \text{ wrap}, \text{glue}]) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{filterp}_2 \sqsubseteq E(\text{ok}(w)) \text{sort}(\tau) \\
& \wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{merge}(\tau) \sqsubseteq \text{cup} \text{sort}(\tau) \wedge \\
& \text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{len}, w)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp} \\
& \langle \text{listf}_1, \text{listf}_2 \text{filterp}_2 \rangle \text{merge}(\tau) \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\text{est}(R)} \\
& \text{a paragraph laid out as a fold that thins the layouts kept at each word}
\end{aligned} \tag{8.5c}$$

circuit	Hinze–Marsden
$ \begin{aligned} & L(\text{Word}) \xrightarrow{\frac{1}{3}} P(L(\text{Word})) \xrightarrow{E(\text{partition fits}(w))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word})) \\ & \frac{\text{partition list}^*(\text{fits } w)}{\text{est}(R)} \end{aligned} $	
$ \begin{aligned} & L(\text{Word}) \xrightarrow{\frac{1}{3}} P(L(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup (\text{glue } (\text{ok } w)))}{\text{est}(R)} \\ & \text{fusion, every word fits on a line by itself.} \end{aligned} $	
$ \begin{aligned} & L(\text{Word}) \xrightarrow{\frac{1}{3}} P(L(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] (\text{ok } w)))}{\text{est}(R)} \\ & \text{the algebra as } (f_1 p_1) \cup (f_2 p_2), p_1 \triangleq 1 \triangleq \text{thinlist.} \end{aligned} $	
$ \begin{aligned} & L(\text{Word}) \xrightarrow{\frac{1}{3}} P(L(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] (\text{ok } w)))}{\text{est}(R)} \\ & \text{the algebra as } (f_1 p_1) \cup (f_2 p_2), p_1 \triangleq 1 \triangleq \text{thinlist.} \end{aligned} $	
$ \begin{aligned} & ([\text{listcp } \langle g_1, g_2 \rangle \text{cat thinlist}(Q)] \text{minlist}(R)) \\ & \text{binary thinning theorem, at } P \triangleq T \text{ with merge } \tau = \text{cat } (\text{merge}(\tau) = \\ & \sqsubseteq \text{cat }), Q \text{ from } F(Q(\text{len}, w)) [\text{wrap wrap}, \text{new}] \sqsubseteq [\text{wrap} \\ & \text{wrap}, \text{new}] Q(\text{len}, w), (\forall a. 0 \leq \text{len}(a)) \\ & \Rightarrow F(Q(\text{len}, w)) [\text{wrap wrap}, \text{glue}] \text{ok}(w) \sqsubseteq [\text{wrap} \\ & \text{wrap}, \text{glue}] \text{ok}(w) Q(\text{len}, w) \end{aligned} $	
$ \begin{aligned} & = ([\text{start}, \text{cpr } \langle h_1, h_2 \rangle \text{cat thinlist}(Q)] \text{minlist}(R)) \\ & \text{listcp} = \text{wrap} + \text{cpr}, g_i \text{ along the coproduct } \triangleq \text{partition} \end{aligned} $	

§8.6. Bitonic tours

(8.6a)

$FA = (\text{City} \times \text{City}) + (\text{City} \times A)$, the base functor of cons-lists of length at least two; $\text{listcp} = \text{wrap} + \text{cpr}$; $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$, neither positive nor symmetric.

$\text{start } (a, b) = ([a, b], [a, b])$, $\text{dropl } (a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$, $\text{dropr } (a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$, $\text{tour} \triangleq ([\text{start}, \text{dropl } \cup \text{dropr}])$.

$\text{cost } (xs, ys) = \text{outcost } xs + \text{incost } ys$, $\text{outcost } [a_0, \dots, a_n] = \text{tc } (a_0, a_1) + \dots + \text{tc } (a_{n-1}, a_n)$, $\text{incost } [a_0, \dots, a_n] = \text{tc } (a_1, a_0) + \dots + \text{tc } (a_n, a_{n-1})$.

$\text{next} \triangleq \text{tail head}$, $\text{next2} \triangleq \text{next} \times \text{next}$, $\text{head2} \triangleq \text{head} \times \text{head}$, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $Q \triangleq R \circ (\text{next2 next2}^\circ) \circ (\text{head2 head2}^\circ)$, $P \triangleq T$, $g_1 \triangleq \text{list}([\text{start}, \text{dropl}])$, $g_2 \triangleq \text{list}([\text{start}, \text{dropr}])$.

(8.6b)

the law	what it says
1 $(1 \times R) \text{dropl} \sqsubseteq \text{dropl } R$ FALSE $(1 \times R) \text{dropr} \sqsubseteq \text{dropr } R$ FALSE	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
2 $F(Q) [\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] Q$ $F(Q) [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] Q$,	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input

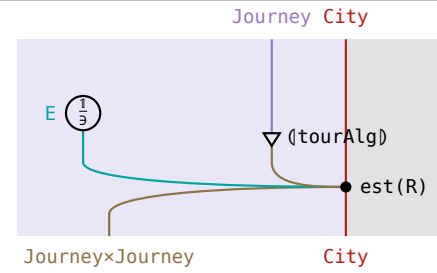
$$\begin{aligned}
& (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{start}, \text{dropl}] \sqcap [\text{start}, \text{dropl}]^\circ) g_1 \sqsubseteq \text{P}([\text{start}, \text{dropl}]) \text{sort}(\tau) \wedge \text{sortF}([\text{start}, \text{dropr}] \sqcap [\text{start}, \text{dropr}]^\circ) g_2 \sqsubseteq \text{P}([\text{start}, \text{dropr}]) \\
& \text{sort}(\tau) \wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{cat} \sqsubseteq \text{cup} \text{sort}(\tau) \wedge \\
& \text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Q) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp}(g_1, g_2) \text{cat} \\
& \text{thinlist}) \text{minlist} \sqsubseteq \frac{(\text{tourAlg})}{3} \text{est}(R)
\end{aligned} \tag{8.6c}$$

a least-cost bitonic tour, as a fold that thins the tours kept at each city

circuit

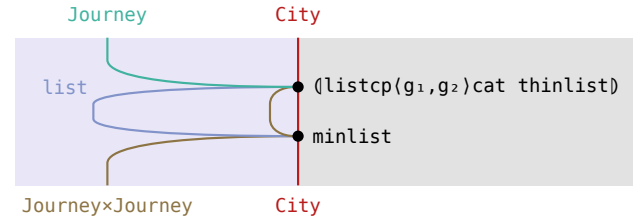
Hinze–Marsden

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{3}} \text{P}(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E((\text{tourAlg}))} \text{P}(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\
& \frac{\text{tour}}{3} \text{est}(R)
\end{aligned}$$



$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{3}} \text{P}(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}])} \text{P}(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\
& = \frac{([\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}])}{3} \text{est}(R) \\
& \text{tour} \triangleq ([\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}]) \triangleq \text{tour}
\end{aligned}$$

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{3}} \text{P}(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}])} \text{P}(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\
& \text{listcp}(g_1, g_2) \text{cat thinlist}(Q) \text{minlist}(R) \\
& \text{binary thinning theorem, at } P \triangleq T \text{ with merge } \tau = \text{cat}, Q \text{ from } F(Q) \\
& [\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] Q, F(Q) [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] Q
\end{aligned}$$



$$\begin{aligned}
& ([\text{start wrap}, \text{cpr}(\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\
& \text{thinlist}(Q)] \text{minlist}(R) \\
& = \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\
& \text{quadratic, two tours added per step}
\end{aligned}$$